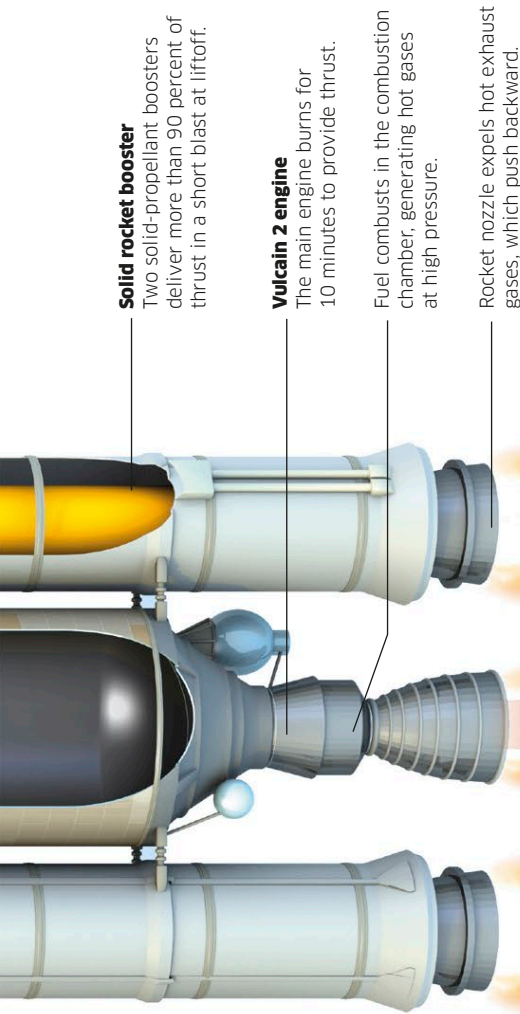




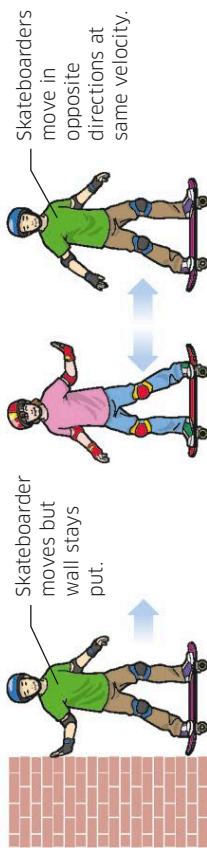
WEIGHT

Newton's three laws at work

A rocket taking off shows Newton's laws in action. Before liftoff, the rocket's enormous weight—the result of gravity pulling it down toward Earth—is balanced by the upward force of the launch pad so it remains stationary (first law). Fuel combustion in the engines creates thrust, propelling the rocket forward (second law). The thrust that pushes the rocket forward is a reaction to the hot exhaust gases pushing backward (third law).

Third law of motion

Forces come in pairs, and any object will react to a force applied to it. The force of reaction is equal and acts in an opposite direction to the force that produces it. If one object is immobile, then the other will move. If both objects can move, then the object with less mass will accelerate more than the other. Every action has an equal and opposite reaction.



Action

If a skateboarder pushes a wall, the wall pushes back with a reaction force that causes the skater to roll away from it.

Reaction

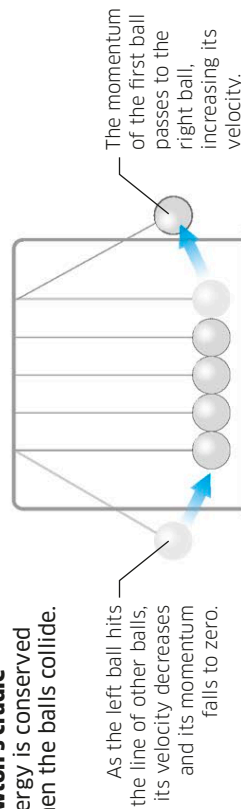
If one skateboarder pushes another, action and reaction cause both skaters to roll away from each other.

Momentum

A moving object keeps moving because it has momentum. It will keep moving until a force stops it. However, when it collides with another object, momentum will be transferred to the second object.

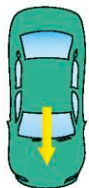
Newton's cradle

Energy is conserved when the balls collide.



Relative velocity

The velocity of an object is its speed in a particular direction. Two objects traveling at the same speed but in opposite directions, or at different speeds in the same direction, have different velocities.



Same direction, same speed

The relative velocity of the two cars is 0 mph (0 km/h).

Same direction, different speeds

The relative velocity of the two cars is 10 mph (15 km/h).

Opposite direction, same speed

The relative velocity of cars on a collision course is 60 mph (100 km/h).

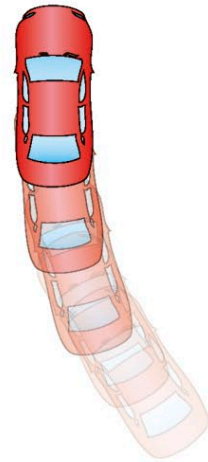
Velocity, speed, and acceleration

Speed is a measure of the rate at which a distance is covered. Velocity is not the same as speed; it measures direction as well as speed of movement. Acceleration measures the rate of change of velocity. Speeding up, turning, and slowing down are all acceleration.



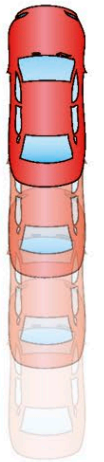
Increasing speed

When a force is applied to an object, its speed increases—it accelerates.



Changing direction

When an object changes direction, its velocity changes. This is also a type of acceleration.



Decreasing speed

When a force slows a moving object down, its speed decreases—it decelerates, or accelerates negatively.